DEVOPS

FROM IBM PERSPECTIVE

By:-

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WHAT IS DEVOPS ??

- DevOps is not a tool, it's a methodology.
- It aims to synchronize & automate the task of Development & Operations Team[1].
- It is the foundation of Rapid Application Development.

WHY DO WE NEED DEVOPS ??

- DevOps promotes coordination and collaboration by leveraging certain tools and encouraging some practices[3].
- It streamlines the agile development lifecycle by introducing small code changes, that are integrated, tested, and deployed to production[3].
- Enable teams to focus on risks by assessing their true impact and contextual relevance[2].

THE PILLARS OF IBM'S FRAMEWORK

Culture

(Collaboration & shared Responsibility)

Process

(Streamline Workflow & Automation)

Tools

Feedback

(Continuous Improvement)

Analysis

(Tracking Performance)

The DevOps Lifecycle

THE DEVOPS LIFECYCLE

Planning

- Features & functionalities are defined in this phase.
- This phase results in structured backlog of features, improvements, and bug fixes.

Coding

- Teams develop new features via practices like TDD, pair programming, and peer code reviews.
- Code is built and tested on local machines before integrating into the CD pipeline.

Building

- Code changes are integrated into the main codebase for automated steps such as merging, testing, compilation, and packaging.
- The resulting build output is stored in a binary repository which holds libraries and executables.

Testing

- Testing is automated to ensure code quality and validating application requirements.
- Continuous, shift-left testing enables faster issue detection and quicker resolution.

THE DEVOPS LIFECYCLE

Release

- This stage performs final checks to ensure application meets quality, security & compliance before production.
- Deployment is automated, including servers & database via IaC.

Deploy

- The application is pushed to production making it accessible to users.
- Rollouts are often gradual, starting with a small user base before full deployment.

Operate

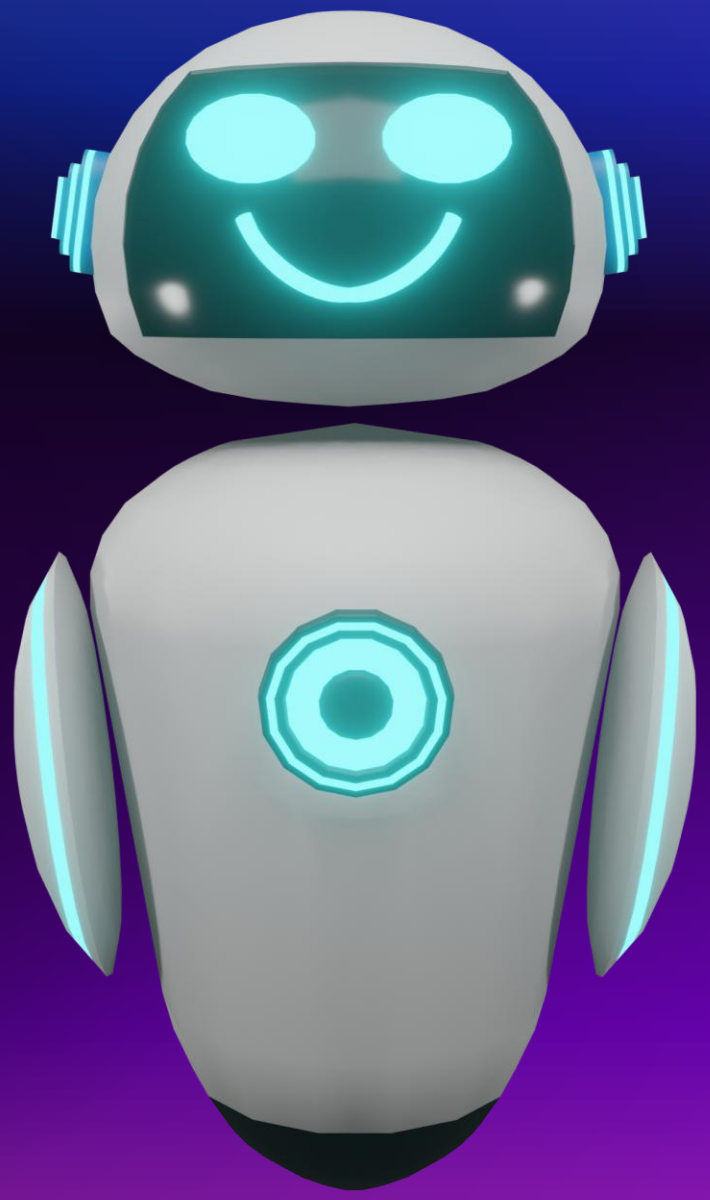
- The application is continuously monitored to ensure features perform reliably & services remain uninterrupted.
- Automated tools are used to optimize operations across infrastructure.

Monitor

- User feedback & system data are continuously analyze to assess application effectiveness.

DEVOPS + AI

- AI can predict & process bottlenecks before they happen.
- AI identify security vulnerabilities in applications & infrastructure, detect potential threats such as a data breach or supply chain attack in real time[1].
- It identify hidden bugs & performance anomalies as well as flag issues that require attention.
- It identify security vulnerabilities that were unintentionally created during a software development.



BENEFITS OF DEVOPS

- Improves collaboration by synchronization development & operations teams[1].
- Enables faster releases via CI/CD practices.
- Ensures stability & reliability of the systems through continuous testing & monitoring[1].
- Faster scaling by automating development & deployment processes.
- Ensures security in the entire lifecycle to maintain compliance and reduce risks.

DEVOPS TOOLS *BY* IBM

IBM Instana

Provide observability in CI/CD with proactive monitoring & early issue detection[1].

IBM Concert

Enhance security by enabling vulnerability management into IDE[1].

Terraform

Enables provisioning & management of infrastructure via IaC[1].

REFERENCES

S.No	Title	Insights
1	The 2026 Guide to DevOps	Integrates development & operations with automation to accelerate reliable software delivery.
2	IBM DevOps Solutions	Modern DevOps uses AI-driven automation, observability & security to reduce risks and deliver faster.
3	UPES DevOps Articulate Lecture	DevOps emphasizes continuous integration, collaboration, and containerization to streamline and standardize the software lifecycle.

DEBUG YOUR DOUBTS

Q1. How Does IBM's Devops Approach Differ From General Devops Practices?

IBM focuses on AI-driven devops practices & enhancing automation, predictive monitoring as well as scalability beyond traditional devops.

Q2. Why is continuous testing important in the DevOps lifecycle?

Continuous testing ensures early bug detection, improving code quality & reducing failures in production.

Q3. How does AI improve DevOps processes?

AI predict bottlenecks, detect vulnerabilities & anomalies in real time, making the system more efficient.

THANK YOU

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